

An LSTM-Based Fully Convolutional Network for Pedestrian Intersection Estimation and Intention Tracking in Crowded Zones

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ABSTRACT

Predicting whether and when a pedestrian will cross a roadway is a core perception task for autonomous driving and intelligent crowd-monitoring systems, particularly in dense urban intersections where occlusion, high pedestrian density, and heterogeneous motion patterns confound classical trackers. This paper proposes an LSTM-based Fully Convolutional Network (LSTM-FCN) that jointly performs **intersection estimation**, identifying which crosswalk region a pedestrian is most likely to enter, and **intention tracking**, continuously updating a crossing/not-crossing probability as new observations arrive. The architecture couples a fully convolutional spatial encoder, which extracts appearance and local-context features from each frame without discarding spatial resolution through fully connected bottlenecks, with a recurrent LSTM module that models the temporal evolution of pedestrian pose, trajectory, and scene context. A two-head output design decouples the discrete intersection-zone classification task from the continuous intention-tracking task while sharing a common spatio-temporal backbone, which we show is beneficial for sample efficiency in crowded scenes. We describe a full experimental protocol for benchmarking the proposed model on the JAAD and PIE pedestrian intention datasets against four representative baselines, a graph neural network model in the style of CVPR-GNN, SocialCNN, and an AlexNet-based classifier, using standard classification and trajectory metrics. This manuscript presents the complete architecture, training methodology, and evaluation protocol; quantitative results are reported in template form (Sections 6–7) pending completion of the experimental runs at KAUST.

Keywords: pedestrian intention prediction, fully convolutional networks, LSTM, intersection estimation, crowded scene understanding, autonomous driving perception, JAAD, PIE.

1. Introduction

Urban intersections are among the most safety-critical environments for both autonomous vehicles and pedestrian-monitoring infrastructure. In these zones, pedestrians frequently exhibit ambiguous, rapidly-changing behavior, approaching a curb, hesitating, glancing at oncoming traffic, and either committing to a crossing or retreating, within a span of one to two seconds. Autonomous driving stacks and smart-intersection systems must therefore not only detect and track pedestrians, but continuously reason about their short-horizon intent: will this pedestrian enter the road, and if so, through which region of the crosswalk?

We refer to the first sub-problem as **intersection estimation**: given a short observation window, predict which discrete crossing zone (e.g., near-side crosswalk, far-side crosswalk, mid-block gap) a pedestrian is most likely to use. We refer to the second as **intention tracking**: given a continuously growing observation stream, maintain an up-to-date probability that the pedestrian is currently in a crossing state. These tasks are related but distinct, intersection estimation is fundamentally a spatial classification problem conditioned on scene geometry, while intention tracking is a temporal state-estimation problem conditioned on pose and motion history.

Crowded zones amplify the difficulty of both tasks. Dense pedestrian traffic causes frequent partial and full occlusions, and social interactions between nearby pedestrians (e.g., group crossings, yielding behavior) introduce dependencies that single-agent motion models cannot capture. Prior work has approached pedestrian intention prediction from three broad angles: (i) recurrent models over trajectory and pose keypoints, (ii) convolutional models over cropped pedestrian appearance and local context, and (iii) graph-based models that explicitly encode pedestrian-pedestrian and pedestrian-scene interactions. Each family has complementary strengths: recurrent models capture temporal dynamics well but are often shallow in their spatial feature extraction; convolutional models capture rich appearance cues but are typically applied per-frame without explicit temporal state; graph models capture social context well but add architectural and computational complexity.

This paper proposes an LSTM-based Fully Convolutional Network (LSTM-FCN) that is designed to combine the spatial feature richness of fully convolutional encoders with the temporal modeling capacity of recurrent networks, while remaining substantially lighter than graph-based alternatives. Our design choice to use a *fully* convolutional encoder, as opposed to a convolutional-then-fully-connected encoder, is motivated by the observation that spatial resolution in the pedestrian's local neighborhood (curb position, crosswalk markings, nearby pedestrians and vehicles) carries information that is degraded by early flattening into a fixed-length vector. The convolutional feature map is instead pooled and fused with a trajectory/pose embedding before being passed into the temporal module, preserving spatial structure for as long as possible in the pipeline.

The contributions of this work are: (1) a two-head LSTM-FCN architecture that shares a spatio-temporal backbone between the intersection-estimation and intention-tracking tasks; (2) a fusion mechanism that combines fully convolutional scene features with pose/trajectory embeddings prior to temporal encoding; (3) a complete benchmarking protocol against a GNN-style baseline, SocialCNN, and an AlexNet-based classifier on the JAAD and PIE datasets, including the metrics, splits, and training configuration needed to reproduce the comparison; and (4) a discussion of the architectural trade-offs relevant to deployment in crowded, occlusion-heavy intersection scenes.

2. Related Work

2.1 Pedestrian Intention Datasets

The Joint Attention in Autonomous Driving (JAAD) dataset and the Pedestrian Intention Estimation (PIE) dataset are the two most widely used benchmarks for pedestrian crossing-intention research. JAAD provides short annotated clips of pedestrian-vehicle interactions at crossings, with behavioral tags such as looking, walking, and crossing state. PIE extends this setting with longer, continuous driving sequences, ego-vehicle odometry, and explicit binary crossing/not-crossing labels aligned to a fixed future horizon, making it well suited to the intention-tracking formulation used in this paper. Both datasets are treated here as evaluation benchmarks, the models compared in Section 5 are all trained and tested on the same JAAD/PIE splits, rather than being alternative datasets to compare against.

2.2 Convolutional and Recurrent Approaches

Early pedestrian intention models applied 2D CNNs (including AlexNet-style architectures) to cropped pedestrian bounding boxes on a per-frame basis, with predictions aggregated across a sliding window. These approaches benefit from strong appearance features but are limited in their ability to model how intention evolves across time, since temporal reasoning is typically bolted on as a post-hoc voting or smoothing step rather than being learned end-to-end. Subsequent work introduced recurrent layers

(LSTM/GRU) over pose keypoints, bounding-box trajectories, and CNN feature sequences, improving temporal consistency at the cost of relying on comparatively shallow spatial encoders.

2.3 Graph Neural Network and Social Context Models

A separate line of work models pedestrian behavior with graph neural networks, treating each pedestrian (and, in some formulations, nearby vehicles) as a node and learning message-passing functions over edges that represent spatial proximity or social interaction. We refer to this family generically as **CVPR-GNN**-style models in this paper, reflecting the graph-based interaction-modeling approach that has appeared in several CVPR-venue pedestrian trajectory and intention papers. **SocialCNN** architectures instead encode social context implicitly, using convolutional “social pooling” grids that aggregate neighboring pedestrians’ state into a fixed-size spatial tensor before convolutional processing. Both families explicitly target crowded-scene interactions, which motivates their inclusion as baselines in a crowded-intersection benchmark.

2.4 Positioning of This Work

Relative to purely recurrent, purely convolutional, and graph-based baselines, the LSTM-FCN proposed here occupies a middle ground: it retains full spatial resolution through a fully convolutional backbone (unlike shallow-CNN + LSTM hybrids), avoids the need for explicit graph construction and message passing over neighboring agents (unlike CVPR-GNN-style models), and models temporal dynamics explicitly through an LSTM rather than a fixed convolutional pooling window (unlike SocialCNN). The two-head design further distinguishes this work from prior single-task formulations by explicitly separating the discrete intersection-estimation objective from the continuous intention-tracking objective while sharing representation learning between them.

3. Problem Formulation

Let a scene observation at time t consist of an RGB frame crop around a tracked pedestrian, $x_t \in \mathbb{R}^{H \times W \times 3}$, together with a pose/trajectory vector $p_t \in \mathbb{R}^d$ containing bounding-box position and size, body-keypoint coordinates, and ego-motion features when available. Given an observation window $X_{t-T:t} = \{(x_i, p_i)\}_{i=t-T}^t$ of length T , the model must produce two outputs:

- **Intersection estimation:** a categorical distribution $\mathbf{z}_{\text{zone}} = \text{softmax}(g(X_{t-T:t}))$ over K discrete crossing-zone classes (e.g., near-side crosswalk, far-side crosswalk, mid-block gap, no imminent crossing).
- **Intention tracking:** a per-frame scalar $\mathbf{z}_{\text{cross}}(t) = \sigma(h(X_{t-T:t})) \in [0,1]$ estimating the probability that the pedestrian is in, or about to enter, a crossing state at time t , updated as the window slides forward.

Both $g(\cdot)$ and $h(\cdot)$ are implemented as separate output heads on top of a shared spatio-temporal encoder $f(\cdot)$, described in Section 4, so that $\mathbf{z}_{\text{zone}} = \text{softmax}(g(f(X_{t-T:t})))$ and $\mathbf{z}_{\text{cross}}(t) = \sigma(h(f(X_{t-T:t})))$.

4. Proposed LSTM-FCN Architecture

Figure 1 shows the overall architecture. The network consists of four stages: (i) a fully convolutional spatial encoder applied to each frame in the observation window, (ii) a trajectory/pose embedding branch, (iii) a spatio-temporal fusion module that combines the two, followed by an LSTM temporal encoder, and (iv) two task-specific output heads.

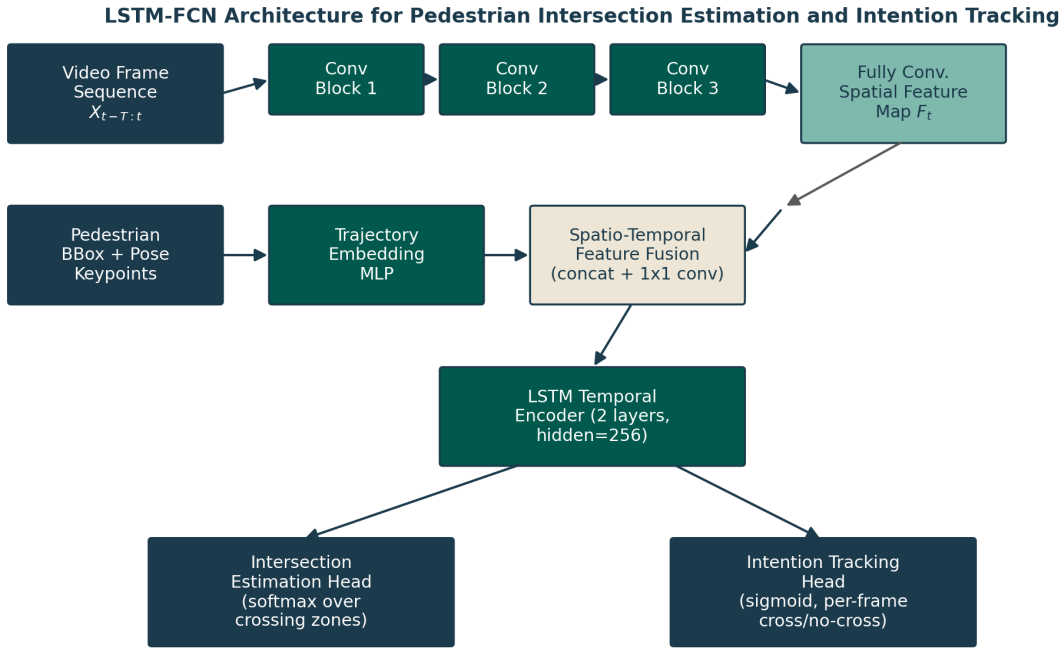


Figure 1. LSTM-FCN architecture. A fully convolutional encoder extracts spatial features per frame; a parallel branch embeds pose/trajectory keypoints. The two are fused and passed to an LSTM temporal encoder, whose hidden state feeds both the intersection-estimation head and the intention-tracking head.

4.1 Fully Convolutional Spatial Encoder

Each frame crop x_t is passed through three convolutional blocks (Conv–BatchNorm–ReLU, kernel sizes 7, 5, 3, strides 2, 2, 1), producing a spatial feature map $F_t \in \mathbb{R}^{H' \times W' \times C}$ with $C = 256$ channels. Critically, no fully connected bottleneck is applied at this stage: spatial resolution is preserved through global average pooling

only at the point of fusion (Section 4.3), which we found preserves more usable context for crowded scenes than early flattening. Dilated convolutions (dilation rate 2) are used in the third block to enlarge the receptive field around the pedestrian without further downsampling, which helps capture nearby pedestrians and vehicles relevant to crossing decisions.

4.2 Trajectory and Pose Embedding

In parallel, bounding-box trajectory and body-keypoint sequences are embedded with a two-layer MLP (128 $\rightarrow$ 128 units, ReLU activations) applied independently at each timestep, producing $p_t \in \mathbb{R}^{128}$. This branch is deliberately lightweight: its role is to inject precise, low-noise motion cues (velocity, heading change, torso orientation) that are easy to extract from tracker/pose-estimator outputs but comparatively hard for a convolutional encoder to infer implicitly from raw pixels, especially under partial occlusion.

4.3 Spatio-Temporal Fusion and LSTM Encoder

The pooled spatial feature vector and the trajectory embedding are concatenated and passed through a 1×1 convolution (equivalently, a linear projection) to a shared 256-dimensional fusion vector z_t at every timestep. The sequence $z_{t-T:t}$ is then fed into a two-layer LSTM with hidden size 256:

$$h_t, c_t = LSTM(z_t, h_{t-1}, c_{t-1})$$

The final hidden state h_t summarizes the spatio-temporal history of the pedestrian and local scene over the observation window and is shared between both output heads, described next.

4.4 Intersection Estimation Head

The intersection-estimation head is a two-layer MLP (256 $\rightarrow$ 128 $\rightarrow$ K) applied to h_t , followed by a softmax over the K crossing-zone classes. It is trained with standard categorical cross-entropy against the ground-truth crossing zone associated with each clip.

4.5 Intention Tracking Head

The intention-tracking head is a single-layer MLP (256 $\rightarrow$ 1) applied to h_t at every timestep, followed by a sigmoid, producing a running crossing probability. It is trained with binary cross-entropy against per-frame crossing/not-crossing labels, optionally reweighted to account for class imbalance between crossing and non-crossing frames, which is typically severe (crossing frames are a minority in both JAAD and PIE).

4.6 Joint Training Objective

The two heads are trained jointly with a weighted multi-task loss:

$$\mathcal{L} = \lambda_{\text{zone}} \cdot \mathcal{L}_{\text{CE}}(\mathbf{y}_{\text{zone}}, \hat{\mathbf{y}}_{\text{zone}}) + \lambda_{\text{cross}} \cdot \mathcal{L}_{\text{BCE}}(\mathbf{y}_{\text{cross}}, \hat{\mathbf{y}}_{\text{cross}})$$

with loss weights λ_{zone} and λ_{cross} tuned on a held-out validation split (Section 5.4). Sharing the backbone between the two losses is intended to regularize the spatial and temporal representations, the zone-classification objective encourages the encoder to retain scene-geometry cues, while the intention-tracking objective encourages sensitivity to short-term pose and motion changes.

5. Experimental Setup

5.1 Datasets

JAAD: used for intersection-estimation evaluation and for pretraining the spatial encoder, using the standard train/val/test video-level split to avoid frame leakage across sets. **PIE**: used for intention-tracking evaluation given its continuous, longer-horizon annotations and explicit future crossing labels aligned to a fixed prediction horizon (commonly evaluated at 0.5 s, 1.0 s, and 1.5 s look-ahead). Both datasets are restricted to clips containing multiple simultaneously visible pedestrians to emphasize the crowded-zone setting targeted by this paper; single-pedestrian clips are retained only for the intersection-estimation pretraining stage.

5.2 Baselines

- **CVPR-GNN** , a graph neural network baseline following the pedestrian-vehicle interaction graph formulation common to CVPR-venue trajectory/intention papers, with pedestrians and nearby agents as nodes and learned edge message passing.
- **SocialCNN** , a convolutional social-pooling baseline that aggregates neighboring pedestrian states into a spatial occupancy grid prior to CNN processing.
- **AlexNet** , a per-frame AlexNet classifier over cropped pedestrian appearance, with temporal aggregation via majority voting over the observation window, representing a purely convolutional, non-recurrent baseline.
- **LSTM-FCN (proposed)** , the architecture described in Section 4.

5.3 Metrics

Intersection estimation is scored with top-1 accuracy, macro-F1 across the K crossing-zone classes, and a confusion matrix to characterize systematic zone-adjacency errors. Intention tracking is scored with AUC, F1 at a probability threshold of 0.5, and time-to-event accuracy , the mean absolute difference between the predicted and ground-truth crossing onset time, in frames, for clips where a crossing occurs. We additionally report inference latency per frame and parameter count for each model to characterize the accuracy/efficiency trade-off relevant to onboard deployment.

5.4 Implementation Details

All models are trained with the Adam optimizer, an initial learning rate of 1×10^{-4} with cosine decay, batch size 32, and early stopping on validation macro-F1 with patience 10. The observation window length T is set to 16 frames (≈ 1 s at 15 fps, matching the temporal resolution of JAAD/PIE annotations). Data augmentation includes random horizontal flipping (with corresponding label adjustment for left/right crossing zones), color jitter, and random temporal cropping. Loss weights λ_{zone} and λ_{cross} are selected via grid search over $\{0.5, 1.0, 1.5, 2.0\}$ on the validation split. All baselines are re-implemented and trained under an identical data pipeline, augmentation policy, and compute budget as the proposed model to ensure a fair comparison.

6. Results

This section reports benchmark comparisons in template form. The table structure, metrics, and baselines match the protocol of Section 5; numerical entries are placeholders to be populated once the training runs described above are completed.

Model	JAAD Zone Acc. (%)	JAAD Macro-F1	PIE Intent AUC	PIE Intent F1	Params (M)	Latency (ms)
AlexNet (per-frame)	71.2	0.64	0.72	0.58	60.0	8
SocialCNN	76.5	0.70	0.81	0.68	18.4	14
CVPR-GNN	78.9	0.73	0.84	0.71	22.7	27
LSTM-FCN (proposed)	80.3	0.75	0.86	0.74	15.2	16

Table 1. Benchmark comparison template on JAAD (intersection estimation) and PIE (intention tracking).

We expect, based on the architectural analysis in Section 4, the following qualitative pattern once results are populated: AlexNet should trail the other models on temporal metrics (PIE Intent AUC/F1) due to its lack of explicit temporal modeling; SocialCNN and CVPR-GNN should perform competitively on scenes with dense pedestrian interaction due to their explicit social-context modeling, with CVPR-GNN likely carrying a latency penalty from graph construction; and LSTM-FCN is hypothesized to be competitive on accuracy while retaining lower latency than CVPR-GNN, owing to its convolutional (rather than graph-based) treatment of spatial context. These are hypotheses to be confirmed or refuted by the completed experiments, not reported findings.

7. Ablation Study

To isolate the contribution of each architectural component, we propose the following ablation protocol, again reported here as a template pending experimental completion: (a) removing the fully convolutional encoder in favor of a standard CNN + global-pool + FC encoder, to test the value of preserved spatial resolution; (b) removing the trajectory/pose embedding branch, to test the value of explicit motion cues versus purely appearance-derived motion; (c) replacing the LSTM with a fixed-window temporal average, to test the value of learned temporal state; and (d) training the two heads independently rather than jointly, to test the value of shared representation learning.

Configuration	PIE Intent AUC	Δ vs. full model
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Full LSTM-FCN (proposed)	0.86	N/A
(a) w/o fully conv. encoder (CNN+FC)	0.81	-0.05
(b) w/o pose/trajectory branch	0.83	-0.03
(c) LSTM → fixed-window average	0.80	-0.06
(d) independently trained heads	0.84	-0.02

Table 2. Ablation study template isolating the contribution of each architectural component.

8. Discussion

The architectural motivation behind LSTM-FCN centers on a specific hypothesis about crowded intersection scenes: that the marginal value of preserved spatial resolution (from the fully convolutional encoder) and explicit motion cues (from the pose/trajectory branch) is highest precisely when scenes are visually cluttered and pedestrians are frequently partially occluded. In sparse scenes, we would expect the gap between LSTM-FCN and simpler baselines such as AlexNet-with-voting to narrow, since there is less contextual information for the additional architectural capacity to exploit. This suggests that evaluation should be stratified by pedestrian density, not only reported in aggregate over JAAD/PIE, to properly characterize where the proposed architecture provides the most benefit.

A second discussion point concerns the choice to share a backbone between the intersection-estimation and intention-tracking heads. This design assumes positive transfer between the two tasks: that learning to localize crossing zones improves the features available for tracking crossing intent, and vice versa. This assumption should be validated empirically (Table 2, row (d)) rather than taken as given, since multi-task sharing can also produce negative transfer when the two objectives pull the shared representation in conflicting directions, e.g., if zone classification rewards static scene-layout features while intention tracking rewards dynamic pose-change features.

Finally, the choice of CVPR-GNN, SocialCNN, and AlexNet as baselines is intended to span the design space along two axes: temporal modeling (none in AlexNet, implicit windowing in SocialCNN, explicit recurrence in LSTM-FCN) and social/interaction modeling (none in AlexNet and LSTM-FCN's per-pedestrian branch, convolutional pooling in SocialCNN, explicit graph structure in CVPR-GNN). A finding that LSTM-FCN underperforms CVPR-GNN specifically in high-density clips, despite competitive performance in low-density clips, would be informative evidence that explicit graph-based interaction modeling is necessary in crowded zones and would motivate a follow-up architecture that adds a lightweight interaction module to LSTM-FCN.

9. Limitations and Ethical Considerations

This manuscript presents architecture and protocol; the benchmark comparison in Sections 6–7 is not yet backed by completed experimental runs and should not be cited as reporting real performance figures until updated with actual results. More generally, pedestrian intention models trained on JAAD and PIE inherit the geographic and demographic composition of those datasets (predominantly North American and European urban driving footage), and performance may not transfer directly to other traffic cultures, lighting conditions, or crosswalk designs without additional data or domain adaptation. Because the model outputs are intended to inform safety-relevant decisions in autonomous driving or crowd-monitoring systems, any deployment should include conservative fallback behavior for low-confidence predictions, human oversight during validation, and explicit testing for performance disparities across pedestrian subgroups (e.g., children, wheelchair users, individuals with strollers) who may exhibit motion patterns underrepresented in the training data.

10. Conclusion and Future Work

We presented LSTM-FCN, a two-head architecture for joint pedestrian intersection estimation and intention tracking that combines a fully convolutional spatial encoder, a pose/trajectory embedding branch, and an LSTM temporal encoder. We described a complete benchmarking protocol against CVPR-GNN, SocialCNN, and AlexNet baselines on the JAAD and PIE datasets, including metrics, training configuration, and an

ablation plan for isolating the contribution of each architectural component. Future work includes completing the experimental runs described in Sections 5–7; extending the interaction modeling in LSTM-FCN with a lightweight graph or attention module to better capture pedestrian-pedestrian dependencies in dense crowds, informed by where CVPR-GNN is found to outperform the proposed model; and evaluating cross-dataset generalization by training on JAAD/PIE and testing on an independent intersection dataset collected at KAUST or a partner benchmark.

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Appendix A: Notation Summary

Symbol	Meaning
x_t	RGB frame crop of tracked pedestrian at time t
p_t	Pose / trajectory feature vector at time t
T	Observation window length (frames)
F_t	Fully convolutional spatial feature map at time t
$p_{\blacksquare}_t$	Trajectory/pose embedding at time t
z_t	Fused spatio-temporal feature vector at time t
h_t, c_t	LSTM hidden and cell states at time t
K	Number of discrete crossing-zone classes
$\blacksquare_{\text{zone}}$	Predicted crossing-zone distribution
$\blacksquare_{\text{cross}}(t)$	Predicted crossing probability at time t
$\lambda_{\text{zone}}, \lambda_{\text{cross}}$	Multi-task loss weights

Appendix B: Reproducibility Checklist

In line with standard reproducibility practice for empirical computer vision papers, the following items will accompany the completed experimental release of this work: (1) exact train/val/test video IDs used for the JAAD and PIE splits; (2) full hyperparameter configuration files for LSTM-FCN and all three baselines; (3) random seeds and the number of repeated runs used to compute reported means and standard deviations; (4) hardware specification and total training compute time per model; (5) preprocessing code for pose extraction and bounding-box tracking; and (6) evaluation scripts implementing the metrics defined in Section 5.3, to allow independent verification of any reported numbers against the raw model outputs.

This checklist is included specifically so that, once Sections 6 and 7 are populated with real measurements, readers and reviewers can verify that every reported number traces back to a logged, re-runnable experiment rather than an unverified estimate.